

AI Based Tools that Aid Research

AI has been around for a very long time, but the current wave seems to be the biggest and most distinctive because it has brought with it tools that are readily usable. Two of the most complex tasks for a research scholar are to search through volumes of literature and understand the complex contents of research papers. This article presents two tools that make both these tasks easier.



Academic research is a challenging task and requires considerable focus and the ability to handle high levels of cognitive load. There are various phases in academic research that can benefit by using specific tools. The application of proper tools can reduce the academic load manifold and help increase the quality of output.

There are many software available that can assist a scholar in the different phases of research. This article focuses on artificial intelligence or AI based tools. An important part of academic research is formulating the research question. Once you find the right question, things become much clearer. To ask the right question the steps researchers follow are:

- Search existing literature
- Understand the papers
- Find the research gaps
- Construct the question

Searching through literature is not as simple as searching a random query in a search engine. It requires specific skills. Two tools that can help a scholar in surveying the literature effectively and understanding its contents easily are:

- Elicit
- Explainpaper

Do remember, however, that these

tools simply assist scholars, and do not automate the research.

Elicit

Elicit is an AI based research assistant, which is based on the GPT-3 (generative pre-trained transformer, third generation) language model. It can assist scholars in searching through the literature and asking the right questions. It has the ability to find answers from 175 million papers. You can run Elicit on your own papers as well as by uploading them.

Elicit is the product of Ought, a non-profit research lab. It currently focuses on machine learning, and is funded by various organisations including Open Philanthropy and Future of Life Institute.



Figure 1: AI for scholars -- Elicit and Explainpaper